

Stratified EEG Connectivity Analysis

2026-05-31

load demographics & connectivity data

```
demo <- read_csv(file.path(base_dir, "Participants_LEMON.csv")) %>%
  rename(subject = `...1`) %>%
  mutate(
    subject = str_remove(subject, "^sub-"),
    subject = str_trim(subject),

    age_low = as.numeric(str_extract(Age, "\\d+")),
    age_high = as.numeric(str_extract(Age, "\\d+$")),
    age_mid = (age_low + age_high) / 2,

    age_group_main = if_else(age_mid <= 60, "young", "old"),

    age_group_ablation = case_when(age_mid < 35 ~ "young", age_mid > 60 ~ "old",
      TRUE ~ NA_character_)

## New names:
## Rows: 228 Columns: 3
## -- Column specification
## ----- Delimiter: "," chr
## (2): ...1, Age dbl (1): Gender_1=female_2=male
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

read_conn <- function(inv_method) {
  path <- file.path(
    base_dir,
    paste0("connectivity_", inv_method),
    "stats",
    "bootstrap_connectivity_summary.tsv"
  )

  read_tsv(path) %>%
    mutate(
      inverse_method = inv_method,
      subject = str_remove(subject, "^sub-"),
      subject = str_trim(subject)
    )
}

conn <- bind_rows(read_conn("dSPM"), read_conn("eLORETA"))
```

```
## Rows: 864 Columns: 7
## -- Column specification -----
## Delimiter: "\t"
## chr (4): subject, task, band, method
## dbl (3): bootstrap_mean, bootstrap_std, bootstrap_snr
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## Rows: 252 Columns: 7
## -- Column specification -----
## Delimiter: "\t"
## chr (4): subject, task, band, method
## dbl (3): bootstrap_mean, bootstrap_std, bootstrap_snr
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
conn2 <- conn %>% left_join(demo, by = "subject")
```

```
conn2 %>% count(age_group_main)
```

```
## # A tibble: 2 x 2
##   age_group_main      n
##   <chr>          <int>
## 1 old             240
## 2 young           876
```

```
conn2 %>% count(age_group_ablation)
```

```
## # A tibble: 2 x 2
##   age_group_ablation      n
##   <chr>          <int>
## 1 old             240
## 2 young           876
```

```
demo %>%
  filter(age_mid >= 35, age_mid <= 60) %>%
  select(subject, Age, age_mid)
```

```
## # A tibble: 5 x 3
##   subject Age   age_mid
##   <chr>   <chr>   <dbl>
## 1 032378 55-60    57.5
## 2 032391 35-40    37.5
## 3 032463 55-60    57.5
## 4 032482 55-60    57.5
## 5 032486 55-60    57.5
```

```
conn %>%
  semi_join(
    demo %>%
      filter(age_mid >= 35, age_mid <= 60), by = "subject") %>%
  count(subject)
```

```
## # A tibble: 0 x 2
## # i 2 variables: subject <chr>, n <int>
```

Stratified Wilcoxon age tests

```
age_tests_main_stratified <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
    !is.na(age_group_main)
  ) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_main == "young"),
    n_old = sum(age_group_main == "old"),
    mean_young = mean(bootstrap_mean[age_group_main == "young"], na.rm = TRUE),
    mean_old = mean(bootstrap_mean[age_group_main == "old"], na.rm = TRUE),
    p = ifelse(
      n_young > 0 & n_old > 0,
      wilcox.test(bootstrap_mean ~ age_group_main, exact = FALSE)$p.value,
      NA_real_
    ),
    .groups = "drop"
  ) %>%
  group_by(inverse_method, method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)

age_tests_main_stratified
```

```
## # A tibble: 24 x 10
##   inverse_method method   band task n_young n_old mean_young mean_old    p
##   <chr>           <chr>   <chr> <chr>   <int> <int>     <dbl>     <dbl> <dbl>
## 1 eLORETA        imcoh   alpha Eyes~    7     3     0.0245    0.0183 0.0682
## 2 eLORETA        wpli2_de~ alpha Eyes~    8     3     0.129     0.110 0.262
## 3 eLORETA        wpli2_de~ alpha Eyes~    7     3     0.110     0.0831 0.254
## 4 eLORETA        wpli2_de~ beta  Eyes~    8     3     0.0435    0.0498 0.262
## 5 eLORETA        imcoh   alpha Eyes~    8     3     0.0279    0.0240 0.185
## 6 dSPM           wpli2_de~ alpha Eyes~   29     7     0.129     0.124 0.522
## 7 dSPM           wpli2_de~ alpha Eyes~   29     7     0.0991    0.0821 0.263
## 8 dSPM           wpli2_de~ beta  Eyes~   29     7     0.0481    0.0550 0.299
## 9 dSPM           wpli2_de~ beta  Eyes~   29     7     0.0451    0.0537 0.358
## 10 dSPM          wpli2_de~ theta Eyes~   29     7     0.0611    0.0552 0.603
## # i 14 more rows
## # i 1 more variable: p_fdr_stratified <dbl>
```

```
age_tests_ablation_stratified <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
```

```

    !is.na(age_group_ablation)
  ) %>%
group_by(inverse_method, method, band, task) %>%
summarise(
  n_young = sum(age_group_ablation == "young"),
  n_old = sum(age_group_ablation == "old"),
  mean_young = mean(bootstrap_mean[age_group_ablation == "young"], na.rm = TRUE),
  mean_old = mean(bootstrap_mean[age_group_ablation == "old"], na.rm = TRUE),
  p = ifelse(
    n_young > 0 & n_old > 0,
    wilcox.test(bootstrap_mean ~ age_group_ablation, exact = FALSE)$p.value,
    NA_real_
  ),
  .groups = "drop"
) %>%
group_by(inverse_method, method) %>%
mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
ungroup() %>%
arrange(p_fdr_stratified)

age_tests_ablation_stratified

```

```

## # A tibble: 24 x 10
##   inverse_method method   band task n_young n_old mean_young mean_old    p
##   <chr>          <chr>   <chr> <chr> <int> <int>      <dbl>    <dbl> <dbl>
## 1 eLORETA       imcoh    alpha Eyes~     7     3    0.0245    0.0183 0.0682
## 2 eLORETA       wpli2_de~ alpha Eyes~     8     3    0.129     0.110 0.262
## 3 eLORETA       wpli2_de~ alpha Eyes~     7     3    0.110     0.0831 0.254
## 4 eLORETA       wpli2_de~ beta  Eyes~     8     3    0.0435    0.0498 0.262
## 5 eLORETA       imcoh    alpha Eyes~     8     3    0.0279    0.0240 0.185
## 6 dSPM          wpli2_de~ alpha Eyes~    29     7    0.129     0.124 0.522
## 7 dSPM          wpli2_de~ alpha Eyes~    29     7    0.0991    0.0821 0.263
## 8 dSPM          wpli2_de~ beta  Eyes~    29     7    0.0481    0.0550 0.299
## 9 dSPM          wpli2_de~ beta  Eyes~    29     7    0.0451    0.0537 0.358
## 10 dSPM         wpli2_de~ theta Eyes~    29     7    0.0611    0.0552 0.603
## # i 14 more rows
## # i 1 more variable: p_fdr_stratified <dbl>

```

The demographic file contained five subjects between 35 and 60, but none of those subjects were present in the connectivity summary data. The main age split and the ablation split used the same effective connectivity sample.

ks tests

```

ks_age_main <- conn2 %>%
  filter(band %in% bands, method %in% conn_types, !is.na(age_group_main)) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_main == "young"),
    n_old = sum(age_group_main == "old"),
    ks_p = ifelse(

```

```

n_young > 0 & n_old > 0,
ks.test(
  bootstrap_mean[age_group_main == "young"],
  bootstrap_mean[age_group_main == "old"]
)$p.value,
NA_real_
),
.groups = "drop") %>%
group_by(inverse_method, method) %>%
mutate(ks_p_fdr = p.adjust(ks_p, method = "BH")) %>%
ungroup() %>%
arrange(ks_p_fdr)

```

ks_age_main

```

## # A tibble: 24 x 8
##   inverse_method method      band task      n_young n_old ks_p ks_p_fdr
##   <chr>          <chr>      <chr> <chr>      <int> <int> <dbl> <dbl>
## 1 eLORETA      wpli2_debiased alpha EyesClosed      8     3 0.303    0.6
## 2 eLORETA      wpli2_debiased alpha EyesOpen       7     3 0.25     0.6
## 3 eLORETA      wpli2_debiased beta  EyesClosed      8     3 0.121    0.6
## 4 eLORETA      wpli2_debiased beta  EyesOpen       7     3 0.4      0.6
## 5 eLORETA      wpli2_debiased theta EyesClosed      8     3 0.709    0.709
## 6 eLORETA      wpli2_debiased theta EyesOpen       7     3 0.7      0.709
## 7 dSPM         wpli2_debiased beta  EyesClosed     29     7 0.141    0.848
## 8 eLORETA      imcoh         alpha EyesClosed      8     3 0.303    0.85
## 9 eLORETA      imcoh         alpha EyesOpen       7     3 0.167    0.85
## 10 eLORETA     imcoh         beta  EyesClosed      8     3 0.709    0.85
## # i 14 more rows

```

```

ks_age_ablation <- conn2 %>%
  filter(band %in% bands, method %in% conn_types, !is.na(age_group_ablation)) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_ablation == "young"),
    n_old = sum(age_group_ablation == "old"),
    ks_p = ifelse(
      n_young > 0 & n_old > 0,
      ks.test(
        bootstrap_mean[age_group_ablation == "young"],
        bootstrap_mean[age_group_ablation == "old"]
      )$p.value,
      NA_real_
    ),
    .groups = "drop") %>%
  group_by(inverse_method, method) %>%
  mutate(ks_p_fdr = p.adjust(ks_p, method = "BH")) %>%
  ungroup() %>%
  arrange(ks_p_fdr)

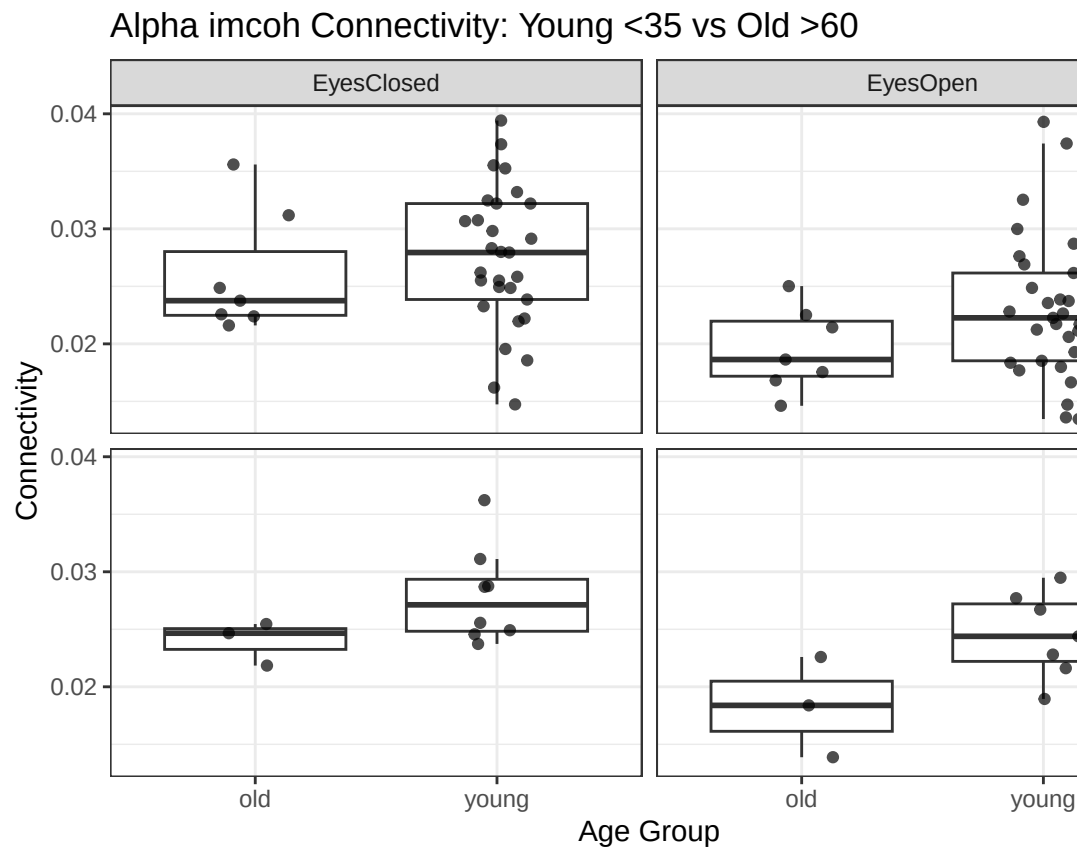
```

ks_age_ablation

```
## # A tibble: 24 x 8
```

```
##      inverse_method method      band task      n_young n_old  ks_p ks_p_fdr
##      <chr>          <chr>      <chr> <chr>      <int> <int> <dbl> <dbl>
##  1 eLORETA         wpli2_debiased alpha EyesClosed      8     3 0.303    0.6
##  2 eLORETA         wpli2_debiased alpha EyesOpen       7     3 0.25     0.6
##  3 eLORETA         wpli2_debiased beta  EyesClosed      8     3 0.121    0.6
##  4 eLORETA         wpli2_debiased beta  EyesOpen       7     3 0.4      0.6
##  5 eLORETA         wpli2_debiased theta EyesClosed      8     3 0.709    0.709
##  6 eLORETA         wpli2_debiased theta EyesOpen       7     3 0.7      0.709
##  7 dSPM            wpli2_debiased beta  EyesClosed     29     7 0.141    0.848
##  8 eLORETA         imcoh          alpha EyesClosed      8     3 0.303    0.85
##  9 eLORETA         imcoh          alpha EyesOpen       7     3 0.167    0.85
## 10 eLORETA         imcoh          beta  EyesClosed      8     3 0.709    0.85
## # i 14 more rows
```

```
ggplot(
  conn2 %>%
    filter(band == "alpha", method == "imcoh", !is.na(age_group_ablation)),
  aes(x = age_group_ablation, y = bootstrap_mean)) +
  geom_boxplot(outlier.shape = NA) + geom_jitter(width = 0.15, alpha = 0.7) +
  facet_grid(inverse_method ~ task) + theme_bw() +
  labs(x = "Age Group", y = "Connectivity",
       title = "Alpha imcoh Connectivity: Young <35 vs Old >60"
  )
```



box plot with jitter points

ANOVA and ANCOVA

```
anova_data <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
    !is.na(age_group_ablation)
  )

anova_model <- lm(
  bootstrap_mean ~ age_group_ablation * band * task * inverse_method * method,
  data = anova_data
)

Anova(anova_model, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: bootstrap_mean
##
## Sum Sq Df F value
## (Intercept) 0.004730 1 16.0795
## age_group_ablation 0.000012 1 0.0394
## band 0.000746 2 1.2679
## task 0.000147 1 0.4997
## inverse_method 0.000008 1 0.0288
## method 0.033946 1 115.3992
## age_group_ablation:band 0.000019 2 0.0327
## age_group_ablation:task 0.000013 1 0.0432
## band:task 0.000187 2 0.3183
## age_group_ablation:inverse_method 0.000010 1 0.0340
## band:inverse_method 0.000035 2 0.0601
## task:inverse_method 0.000001 1 0.0022
## age_group_ablation:method 0.000022 1 0.0760
## band:method 0.007708 2 13.1019
## task:method 0.002252 1 7.6571
## inverse_method:method 0.000176 1 0.5976
## age_group_ablation:band:task 0.000044 2 0.0752
## age_group_ablation:band:inverse_method 0.000011 2 0.0190
## age_group_ablation:task:inverse_method 0.000000 1 0.0001
## band:task:inverse_method 0.000006 2 0.0103
## age_group_ablation:band:method 0.000181 2 0.3068
## age_group_ablation:task:method 0.000158 1 0.5357
## band:task:method 0.002066 2 3.5120
## age_group_ablation:inverse_method:method 0.000125 1 0.4250
## band:inverse_method:method 0.000388 2 0.6593
## task:inverse_method:method 0.000121 1 0.4103
## age_group_ablation:band:task:inverse_method 0.000004 2 0.0063
## age_group_ablation:band:task:method 0.000294 2 0.5004
## age_group_ablation:band:inverse_method:method 0.000183 2 0.3108
## age_group_ablation:task:inverse_method:method 0.000012 1 0.0401
## band:task:inverse_method:method 0.000149 2 0.2531
## age_group_ablation:band:task:inverse_method:method 0.000076 2 0.1298
## Residuals 0.150020 510
```

```
##                                Pr(>F)
## (Intercept)                   6.982e-05 ***
## age_group_ablation            0.84264
## band                          0.28231
## task                          0.47997
## inverse_method                0.86542
## method                        < 2.2e-16 ***
## age_group_ablation:band       0.96787
## age_group_ablation:task       0.83551
## band:task                     0.72754
## age_group_ablation:inverse_method 0.85368
## band:inverse_method          0.94163
## task:inverse_method          0.96293
## age_group_ablation:method     0.78285
## band:method                  2.827e-06 ***
## task:method                  0.00586 **
## inverse_method:method         0.43986
## age_group_ablation:band:task  0.92758
## age_group_ablation:band:inverse_method 0.98115
## age_group_ablation:task:inverse_method 0.99357
## band:task:inverse_method     0.98975
## age_group_ablation:band:method 0.73592
## age_group_ablation:task:method 0.46455
## band:task:method             0.03056 *
## age_group_ablation:inverse_method:method 0.51472
## band:inverse_method:method    0.51765
## task:inverse_method:method    0.52211
## age_group_ablation:band:task:inverse_method 0.99367
## age_group_ablation:band:task:method 0.60661
## age_group_ablation:band:inverse_method:method 0.73299
## age_group_ablation:task:inverse_method:method 0.84143
## band:task:inverse_method:method 0.77647
## age_group_ablation:band:task:inverse_method:method 0.87832
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ancova_model <- lm(
  bootstrap_mean ~ age_mid * band * task * inverse_method * method,
  data = anova_data
)
```

```
Anova(ancova_model, type = 3)
```

```
## Anova Table (Type III tests)
```

```
##
## Response: bootstrap_mean
##
```

	Sum Sq	Df	F value	Pr(>F)
## (Intercept)	0.005586	1	19.1085	1.497e-05 ***
## age_mid	0.000004	1	0.0143	0.9047
## band	0.001106	2	1.8926	0.1517
## task	0.000025	1	0.0841	0.7720
## inverse_method	0.000009	1	0.0314	0.8594
## method	0.036819	1	125.9557	< 2.2e-16 ***


```

## age_mid:band                0.000022    2    0.0372    0.9635
## age_mid:task                0.000021    1    0.0720    0.7886
## band:task                   0.000005    2    0.0092    0.9909
## age_mid:inverse_method      0.000014    1    0.0465    0.8293
## band:inverse_method         0.000010    2    0.0166    0.9835
## task:inverse_method         0.000001    1    0.0033    0.9544
## age_mid:method              0.000000    1    0.0014    0.9700
## band:method                 0.009375    2   16.0360  1.761e-07 ***
## task:method                 0.000454    1    1.5521    0.2134
## inverse_method:method        0.000074    1    0.2538    0.6147
## age_mid:band:task           0.000057    2    0.0980    0.9066
## age_mid:band:inverse_method 0.000014    2    0.0243    0.9760
## age_mid:task:inverse_method 0.000000    1    0.0001    0.9939
## band:task:inverse_method     0.000001    2    0.0020    0.9980
## age_mid:band:method          0.000221    2    0.3779    0.6855
## age_mid:task:method          0.000298    1    1.0189    0.3132
## band:task:method             0.000161    2    0.2761    0.7588
## age_mid:inverse_method:method 0.000186    1    0.6352    0.4258
## band:inverse_method:method   0.000063    2    0.1085    0.8972
## task:inverse_method:method   0.000015    1    0.0504    0.8224
## age_mid:band:task:inverse_method 0.000004    2    0.0068    0.9932
## age_mid:band:task:method      0.000454    2    0.7762    0.4607
## age_mid:band:inverse_method:method 0.000237    2    0.4049    0.6673
## age_mid:task:inverse_method:method 0.000016    1    0.0546    0.8153
## band:task:inverse_method:method 0.000017    2    0.0297    0.9707
## age_mid:band:task:inverse_method:method 0.000079    2    0.1347    0.8740
## Residuals                   0.149082  510
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

dSPM vs eLORETA using paired testing Wilcoxon-Signed rank test

```

dspm_vs_eloreta <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types
  ) %>%
  select(subject, task, band, method, inverse_method, bootstrap_mean) %>%
  pivot_wider(
    names_from = inverse_method,
    values_from = bootstrap_mean
  ) %>%
  filter(!is.na(dSPM), !is.na(eLORETA)) %>%
  group_by(method, band, task) %>%
  summarise(
    mean_dSPM = mean(dSPM, na.rm = TRUE),
    mean_eLORETA = mean(eLORETA, na.rm = TRUE),
    p = wilcox.test(dSPM, eLORETA, paired = TRUE, exact = FALSE)$p.value,
    .groups = "drop"
  ) %>%
  group_by(method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%

```

```
ungroup() %>%
  arrange(p_fdr_stratified)

dspm_vs_eloreta
```

```
## # A tibble: 12 x 7
##   method      band task      mean_dSPM mean_eLORETA      p p_fdr_stratified
##   <chr>      <chr> <chr>      <dbl>      <dbl> <dbl>      <dbl>
## 1 imcoh      theta EyesClos~ 0.0193      0.0197 0.0367      0.220
## 2 imcoh      beta  EyesClos~ 0.0101      0.00985 0.100        0.300
## 3 imcoh      alpha EyesClos~ 0.0271      0.0269 0.563        0.676
## 4 imcoh      alpha EyesOpen 0.0228      0.0227 0.476        0.676
## 5 imcoh      theta EyesOpen 0.0207      0.0209 0.359        0.676
## 6 wpli2_debiased theta EyesClos~ 0.0678      0.0693 0.120        0.718
## 7 imcoh      beta  EyesOpen 0.0103      0.0103 0.919        0.919
## 8 wpli2_debiased alpha EyesClos~ 0.124       0.124 0.965        1
## 9 wpli2_debiased alpha EyesOpen 0.102       0.102 1            1
## 10 wpli2_debiased beta  EyesClos~ 0.0457      0.0452 1            1
## 11 wpli2_debiased beta  EyesOpen 0.0463      0.0460 0.838        1
## 12 wpli2_debiased theta EyesOpen 0.0760      0.0766 0.359        1
```

WPLI-debiased vs IMCOH using paired testing Wilcoxon-Signed rank test

```
wpli_vs_imcoh <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types
  ) %>%
  select(subject, task, band, inverse_method, method, bootstrap_mean) %>%
  pivot_wider(
    names_from = method,
    values_from = bootstrap_mean
  ) %>%
  filter(!is.na(wpli2_debiased), !is.na(imcoh)) %>%
  group_by(inverse_method, band, task) %>%
  summarise(
    mean_wpli = mean(wpli2_debiased, na.rm = TRUE),
    mean_imcoh = mean(imcoh, na.rm = TRUE),
    p = wilcox.test(wpli2_debiased, imcoh, paired = TRUE, exact = FALSE)$p.value,
    .groups = "drop"
  ) %>%
  group_by(inverse_method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)

wpli_vs_imcoh
```

```
## # A tibble: 12 x 7
##   inverse_method band task      mean_wpli mean_imcoh      p p_fdr_stratified
##   <chr>      <chr> <chr>      <dbl>      <dbl> <dbl>      <dbl>
```

## 1	dSPM	alpha EyesClosed	0.128	0.0271	1.75e-7	0.000000175
## 2	dSPM	alpha EyesOpen	0.0958	0.0224	1.75e-7	0.000000175
## 3	dSPM	beta EyesClosed	0.0494	0.0111	1.75e-7	0.000000175
## 4	dSPM	beta EyesOpen	0.0467	0.0106	1.75e-7	0.000000175
## 5	dSPM	theta EyesClosed	0.0600	0.0174	1.75e-7	0.000000175
## 6	dSPM	theta EyesOpen	0.0640	0.0184	1.75e-7	0.000000175
## 7	eLORETA	alpha EyesClosed	0.124	0.0269	3.86e-3	0.00592
## 8	eLORETA	alpha EyesOpen	0.102	0.0227	5.92e-3	0.00592
## 9	eLORETA	beta EyesClosed	0.0452	0.00985	3.86e-3	0.00592
## 10	eLORETA	beta EyesOpen	0.0460	0.0103	5.92e-3	0.00592
## 11	eLORETA	theta EyesClosed	0.0693	0.0197	3.86e-3	0.00592
## 12	eLORETA	theta EyesOpen	0.0766	0.0209	5.92e-3	0.00592

PERMANOVA

```
run_permanova <- function(data, age_var, inv_method) {

  wide_data <- data %>%
    filter(inverse_method == inv_method, band %in% bands, method %in% conn_types,
           !is.na(data[[age_var]])) %>%
    select(subject, age_group = all_of(age_var), band, task, method,
           bootstrap_mean) %>%
    unite(feature, band, task, method, sep = "_") %>%
    pivot_wider(names_from = feature, values_from = bootstrap_mean) %>%
    drop_na()

  print(wide_data %>% count(age_group))

  X <- wide_data %>% select(-subject, -age_group)

  meta <- wide_data %>% select(subject, age_group)

  permanova <- adonis2(X ~ age_group, data = meta, method = "euclidean",
                      permutations = 9999)

  dist_mat <- dist(X)

  disp <- betadisper(dist_mat, meta$age_group)

  permdisp <- permutest(disp, permutations = 9999)

  list(data_counts = wide_data %>% count(age_group), permanova = permanova,
       permdisp = permdisp)
}

permanova_dspm_main <- run_permanova(conn2, age_var = "age_group_main",
                                     inv_method = "dSPM")
```

```
## # A tibble: 2 x 2
##   age_group      n
##   <chr>        <int>
## 1 old          7
```

```
## 2 young      29
```

```
permanova_eloreta_main <- run_permanova(conn2, age_var = "age_group_main",  
                                         inv_method = "eLORETA")
```

```
## # A tibble: 2 x 2  
##   age_group    n  
##   <chr>      <int>  
## 1 old        3  
## 2 young      7
```

```
permanova_dspm_ablation <- run_permanova(conn2, age_var = "age_group_ablation",  
                                           inv_method = "dSPM")
```

```
## # A tibble: 2 x 2  
##   age_group    n  
##   <chr>      <int>  
## 1 old        7  
## 2 young     29
```

```
permanova_eloreta_ablation <- run_permanova(conn2, age_var = "age_group_ablation",  
                                              inv_method = "eLORETA")
```

```
## # A tibble: 2 x 2  
##   age_group    n  
##   <chr>      <int>  
## 1 old        3  
## 2 young      7
```

```
permanova_dspm_main$permanova
```

```
## Permutation test for adonis under reduced model  
## Permutation: free  
## Number of permutations: 9999  
##  
## adonis2(formula = X ~ age_group, data = meta, permutations = 9999, method = "euclidean")  
##           Df SumOfSqs      R2      F Pr(>F)  
## Model      1 0.003031 0.02224 0.7732 0.5011  
## Residual 34 0.133298 0.97776  
## Total     35 0.136329 1.00000
```

```
permanova_dspm_main$permdisp
```

```
##  
## Permutation test for homogeneity of multivariate dispersions  
## Permutation: free  
## Number of permutations: 9999  
##  
## Response: Distances  
##           Df    Sum Sq   Mean Sq      F N.Perm Pr(>F)  
## Groups      1 0.0000079 0.00000787 0.0112  9999 0.9204  
## Residuals 34 0.0238425 0.00070125
```

```
permanova_eloreta_main$permanova
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = X ~ age_group, data = meta, permutations = 9999, method = "euclidean")
##      Df SumOfSqs      R2      F Pr(>F)
## Model    1 0.0028506 0.15964 1.5198 0.2013
## Residual  8 0.0150058 0.84036
## Total    9 0.0178564 1.00000
```

```
permanova_eloreta_main$permdisp
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 9999
##
## Response: Distances
##      Df      Sum Sq    Mean Sq      F N.Perm Pr(>F)
## Groups    1 0.00000895 8.9450e-06 0.0351   9999 0.8632
## Residuals  8 0.00204065 2.5508e-04
```

```
permanova_dspm_ablation$permanova
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = X ~ age_group, data = meta, permutations = 9999, method = "euclidean")
##      Df SumOfSqs      R2      F Pr(>F)
## Model    1 0.003031 0.02224 0.7732 0.5068
## Residual 34 0.133298 0.97776
## Total   35 0.136329 1.00000
```

```
permanova_dspm_ablation$permdisp
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 9999
##
## Response: Distances
##      Df      Sum Sq    Mean Sq      F N.Perm Pr(>F)
## Groups    1 0.0000079 0.00000787 0.0112   9999 0.9193
## Residuals 34 0.0238425 0.00070125
```

```
permanova_eloreta_ablation$permanova
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## adonis2(formula = X ~ age_group, data = meta, permutations = 9999, method = "euclidean")
##      Df SumOfSqs      R2      F Pr(>F)
## Model    1 0.0028506 0.15964 1.5198 0.1998
## Residual  8 0.0150058 0.84036
## Total    9 0.0178564 1.00000
```

```
permanova_eloreta_ablation$permdisp
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 9999
##
## Response: Distances
##      Df      Sum Sq    Mean Sq      F N.Perm Pr(>F)
## Groups    1 0.00000895 8.9450e-06 0.0351   9999 0.8699
## Residuals  8 0.00204065 2.5508e-04
```